

Leo Lin

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EDUCATION

University of Illinois Urbana Champaign Expected May 2027
Bachelor of Science in Computer Engineering Senior Standing | GPA: 3.98/4.00
ECE Visionary Scholarship (top 8 from over 500 ECE majors), IL Engineering Achievement Scholarship

Related/Current Coursework:

Data Structures (C++), Algorithms & Models of Computation, Comp Sci I & II (Java & C++), Obj-Orien Prog (Python)

TECHNICAL SKILLS

Programming Languages: Advanced in Python and C++; Proficient in C and Java
Framework, Tools: Git, Github, React, HTML, Docker, Arduino IDE, Ki-Cad, UART

WORK & RESEARCH EXPERIENCE

Augmented Listening Technology Lab Urbana, IL
Embedded Software Researcher Jan 2023 - Current

- Developed silent robot head turret that creates audio datasets w/ motion, improving upon stationary ones
- Built turret to both nod and turn like humans, an extra degree of freedom from acoustic head simulators on the market
- Coded closed-loop system in C on Arduino w/ induction sensors, improving recentring error to within ± 5 degrees
- Configured TMC motor driver chips w/ UART to reduce noise by $\sim 40\%$ (measured w/ handheld sound meters)
- Designed and presented poster at symposium, to demonstrate turret capabilities to students and professors

Robnux Sunnyvale, CA
Software Engineering Intern May 2024 - Aug 2024

- Responsible for modernizing the company's Robot calibration program (PyQt frontend and C++ backend)
- Moved the outdated local PyQt interface to the web through React and Python, allowing the use of Docker
- Deployed app into Docker environment with real-time file sharing bt. local and virtual machine using bind mounts, removing the need for manually updating the application locally on hundreds of factory computers

OTHER PROJECT HIGHLIGHTS

Tomato Farmers helper robot (HackIllinois hackathon under John Deere Track) Feb 2024

- Made an autonomous car under 48 hours that identifies balls of a certain color and tracks/chases them around
- Revamped code from unsupported Picamera library to Picamera2 using only raw documentation for reference
- Devised a Python openCV color-mask ball-detecting algorithm through new library w/ 90% accuracy

MIT Beaver Works Summer Institute (Autonomous Underwater Vehicles (AUV) project) June 2022 - Aug 2022

- Programmed AUVs to navigate a simulated underwater treasure-hunting map on Linux VMs through remote SSH
- Perfected communication bt. 3 different AUV types, becoming the only team to utilize all vehicles in competition
- Implemented information sharing and movement of vehicles using the MOOS-IvP library in C++

Squirrel Robot Chasing Project Jun 2021 - Sep 2021

- Built this robot using Raspberry Pi, solar panel, chassis, and ultrasonic sensors to address complaints of squirrels
- Trained an AI model for image detection of squirrels using Nanonets API, with up to 270 degrees field of view
- Wrote robot driving and object avoidance algorithm in Python, allowing robot to patrol over two thirds of our backyard

First Tech Challenge Robotics (Team Paragon 11311) San Jose, CA
Robot Driver and Technical Lead Sep 2018 – May 2023

- Tested and debugged robot code to ensure our team had better autonomous performance during tournaments
- Led team to win the Santa Clara qualifier and advanced to NorCal regionals both years as the robot operator